

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q13-0281

Orangedale 138 kV  
September 18, 2014

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0281. The Interconnection Customer proposes a 113.3 MW wind facility interconnecting at the Orangedale 138 kV substation in the Maryland area. The request was received on October 23, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 205 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,510,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                                               | Contingency                                              | Loading (%) | Rating (MVA) | DFAX (%) |
|------------------------------------------------------------------|----------------------------------------------------------|-------------|--------------|----------|
| HIGH RIDGE / N POLE 138 KV                                       | DIXLETA / TABER 345 KV Line                              | 138.7       | 350          | 9.5      |
| DIXLETA / TABER 345 KV line                                      | SHINGLETOWN / RIVA RD 345 KV                             | 127.1       | 1,108        | 21.3     |
| WESTPORT / WHEELABRATOR<br>BALTIMORE REFUSE SUBSTATION<br>138 KV | DIXLETA / TABER 345 KV                                   | 126.9       | 260          | 6.8      |
| WILKENS / HIGHGROVE 138 KV                                       | TABER / DIXLETA SUBSTATION 345 KV                        | 124.4       | 307          | 6.7      |
| HIGH RIDGE / JEROME 138 KV                                       | DIXLETA / TABER 345 KV                                   | 121.9       | 429          | 9.5      |
| GLENANNAN / TABER 345 KV                                         | SIERRA PACIFIC LUMBER<br>SUBSTATION 138/1 kV Transformer | 120.3       | 1,198        | 23.8     |
| LAE CRK / SIERRA PACIFIC LUMBER<br>138 KV                        | GLENANNAN / TABER 345 KV                                 | 119.7       | 445          | 6.2      |
| BELTSVILLE SUBSTATION / BUNKER<br>HILL 138 KV                    | TABER / GLENANNAN 345 KV                                 | 118.3       | 431          | 7.5      |
| BUNKER HILL / NASA 138 KV                                        | TABER / GLENNANAN 345 KV L                               | 115.5       | 474          | 7.5      |
| SUNBRIGHT / FAIRLAND 138 KV                                      | TABER / GLENANNAN 345 KV                                 | 114.1       | 513          | 9.2      |
| WILKENS / WHEELABRATOR<br>BALTIMORE REFUSE 138 KV line           | TABER / DIXLETA 345 KV L                                 | 112.7       | 373          | 6.7      |
| SIERRA PACIFIC LUMBER 138/1kV<br>XFMR                            | TABER / GLENANAN 345 KV                                  | 112.1       | 747          | 8.2      |
| SIERRA PACIFIC LUMBER 345 kv/1 kV<br>XFMR                        | TABER / GLENANNAN 345 KV                                 | 112.1       | 747          | 8.2      |

| Monitored Facility                                | Contingency                  | Loading (%) | Rating (MVA) | DFAX (%) |
|---------------------------------------------------|------------------------------|-------------|--------------|----------|
| SUNBRIGHT / COLESVILLE 138 KV                     | TABER / GLENANNAN 345 KV     | 111.1       | 577          | 9.2      |
| SIUC / FAIRLAND 138 KV                            | TABER / GLEANNAN 345 KV line | 108.8       | 411          | 5.9      |
| DIXLETA / DRUM SUBSTATION 138 KV L                | TABER / DIXLETA 345 KV       | 107.8       | 523          | 6.3      |
| DIXLETA SUBSTATION / NORTH POLE SUBSTATION 138 KV | TABER / DIXLETA 345 KV       | 104.8       | 270          | 5.9      |
| NORH POLE / DIXLETA 138 KV #2                     | DIXLETA / TABER 345 KV       | 104.8       | 270          | 5.9      |
| TABER 345 kV/138 kV XFMR                          | TABER / DIXLETA 345 KV line  | 104.1       | 767          | 6.5      |

### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

## 4. Required Network Upgrades and Cost Estimates

| Facility                                                   | Required Upgrade                            | Cost Estimate | Allocation to Project |
|------------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| N POLE / HIGH RIDGE SUBSTATION 138 KV                      | Rebuild line with higher-capacity conductor | \$3,763,000   | 2% (\$70,000)         |
| DIXLETA / TABER 345 KV                                     | Rebuild line with higher-capacity conductor | \$30,003,000  | 2% (\$499,000)        |
| WESTPORT / WHEELABRATOR BALTIMORE REFUSE SUBSTATION 138 KV | Rebuild line with higher-capacity conductor | \$849,000     | 2% (\$19,000)         |
| WILKENS / HIGHGROVE 138 KV                                 | Rebuild line with higher-capacity conductor | \$905,000     | 2% (\$21,000)         |
| HIGH RIDGE / JEROME 138 KV                                 | Rebuild line with higher-capacity conductor | \$1,602,000   | 2% (\$30,000)         |
| GLENANNAN / TABER 345 KV                                   | Rebuild line with higher-capacity conductor | \$26,875,000  | 2% (\$467,000)        |
| LAE CRK / SIERRA PACIFIC LUMBER 138 KV                     | Rebuild line with higher-capacity conductor | \$2,156,000   | 4% (\$85,000)         |
| BELTSVILLE SUBSTATION / BUNKER HILL 138 KV                 | Rebuild line with higher-capacity conductor | \$929,000     | 2% (\$24,000)         |
| BUNKER HILL / NASA 138 KV                                  | Rebuild line with higher-capacity conductor | \$2,387,000   | 2% (\$61,000)         |
| SUNBRIGHT / FAIRLAND 138 KV                                | Rebuild line with higher-capacity conductor | \$497,000     | 2% (\$9,000)          |
| WILKENS / WHEELABRATOR BALTIMORE REFUSE 138 KV line        | Rebuild line with higher-capacity conductor | \$1,821,000   | 2% (\$42,000)         |
| SIERRA PACIFIC LUMBER 138-1 kV Transformer Bank            | Replace transformer with higher-rated unit  | \$5,273,000   | 2% (\$132,000)        |
| SIERRA PACIFIC LUMBER 345 kV/1 kV XFMR                     | Replace transformer with higher-rated unit  | \$18,540,000  | 2% (\$465,000)        |
| SUNBRIGHT / COLESVILLE 138 KV                              | Rebuild line with higher-capacity conductor | \$1,158,000   | 2% (\$22,000)         |

| Facility                                          | Required Upgrade                           | Cost Estimate | Allocation to Project |
|---------------------------------------------------|--------------------------------------------|---------------|-----------------------|
| SIUC / FAIRLAND 138 KV                            | Reconductor line                           | \$532,000     | 3% (\$14,000)         |
| DIXLETA / DRUM 138 KV                             | Reconductor line                           | \$699,000     | 2% (\$14,000)         |
| DIXLETA SUBSTATION / NORTH POLE SUBSTATION 138 KV | Reconductor line                           | \$386,000     | 3% (\$10,000)         |
| NORH POLE / DIXLETA 138 KV #2                     | Reconductor line                           | \$494,000     | 3% (\$13,000)         |
| TABER 345 kV/138 kV XFMR                          | Replace transformer with higher-rated unit | \$22,900,000  | 2% (\$513,000)        |
|                                                   | Total allocated to project                 |               | \$2,510,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.